CLAIM - Alpha as a 21|30 Closure Gate v38.4

Primary box formula first; ledger links explain where each factor enters.

The formula below is the primary claim, not a later numerical add-on. The rest of the note is the ledger explaining where each factor enters.

Primary claim: alpha_RT is the closure margin

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260}(3 \cdot 30 - \Delta)$$

Expanded correction

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$

Factor map

Factor	Meaning
pi	circular / half-turn geometry scale
30	C30 carrier/readout grid
1260	first full active closure
3*30	ideal three-chain C30 opening
Delta	typed closure loss/correction
3*30 - Delta	effective remaining opening
pi/(30*1260)	carrier x full-closure normalization

Numerical comparison

external check only - not input

alpha_RT	0.0072973525630485
alpha_inv	137.035999201092
CODATA inv	137.035999177(21)
diff inv	2.409e-08

The gate value is computed from the 21|30 ledger. CODATA is displayed only after the computation.

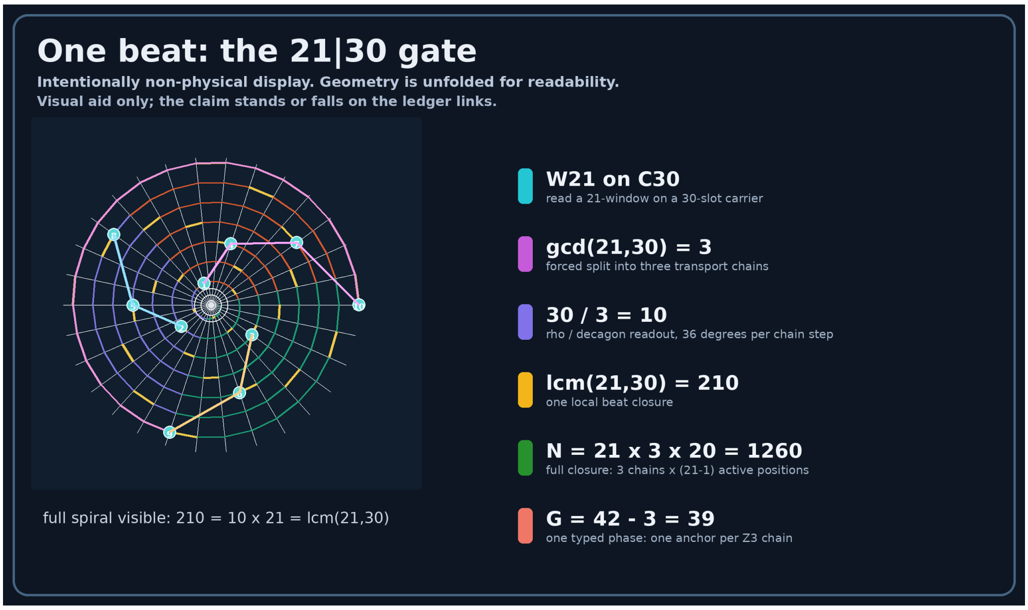
Gate ledger leading to the factors

Link	Consequence
21|30	primitive readout on C30 carrier
3 = gcd(21,30)	three-chain split; source of the 3 in 3*30
10 = 30/3	rho=10 readout scale used in Delta
210 = lcm(21,30)	one beat cell: 7*C30 = 10*21
1260 = 6*210 = 30*42	first full active closure and normalizing scale
G = 42-3 = 3*(14-1) = 39	one A/B-resolved phase reserves one anchor per Z3 chain
D = (21-1)/21 = 20/21	one boundary slot reserved inside carrier-derived 21-domain

Break condition: invalid link, non-unique gate, broken type rule, or equal-or-stricter competing grammar.

Gate detail - carrier-beat and Delta grammar

Six 210-cells split as A/B x Z3; this supports G=39 and the 21-domain for D.



Companion visualizer (not part of the claim): RT_ALPHA_READOUT_GEOMETRY.html. It is an interactive reader aid; the claim stands or falls on ledger links.

Carrier-beat support lemma

Full closure contains six 210-beat cells. With theta(n)=360*n/1260, each 210-boundary advances 60 degrees. A three-arm carrier therefore splits the six cells into two complementary triads.

j	n=j*210	q=n/30	phase	class
0	0	0	0 deg	A
1	210	7	60 deg	B
2	420	14	120 deg	A
3	630	21	180 deg	B
4	840	28	240 deg	A
5	1050	35	300 deg	B

A = {0,14,28}, B = {7,21,35} in C30 index
1260 = 6*210 = (A/B)*Z3*210
G = 42 - 3 = 3*(14 - 1) = 39
AB=2 already counts the complementary phases; G counts one typed phase, so 42-3 not 42-6.
Each phase spans 3 beat cells; each cell has 210/30=7 C30 slots -> 3*7=21.

Correction grammar and type rule

Delta = AB + AB/rho - 1/(rho G) - D/(rho G C S)

Expansion: Delta = 2 + 2/10 - 1/(10*39) - (20/21)/(10*39*30*7).

Type rule: each term is measured in its own domain; numerator schema is AB, AB, 1, D.

Counting measure gives numerator 1; readout-conditioned measure gives numerator D.

Active duty: one typed phase exposes 3*7=21 slots; reserve one -> D=20/21.

alpha_RT^-1 = 137.035999201092; CODATA is external comparison, not input.

Scope and break protocol

Alpha claim only. To refute: break a named gate link, the active-duty derivation, the type rule, or provide an equal-or-stricter competing primitive gate. Free recombinations do not count.